Notes: Matvos & Seru (RFS, 2014)

Resource Allocation within Firms and Financial Market Dislocation: Evidence from Diversified Conglomerates

Contents

1 Big picture

- **Question.** Do firm boundaries mediate financial-sector shocks? More specifically, can diversified firms use internal capital markets to offset dislocations in external capital markets?
- **Economic tension.** Internal capital markets have a **dark side**: headquarters may practice corporate socialism and give too much capital to weak divisions. They also have a **bright side**: headquarters can move funds across divisions without raising costly external finance.
- **Setting.** The paper studies U.S. diversified firms using Compustat segment data from 1980–2006, then uses the estimated model to simulate the 2007–2010 financial-crisis period.
- **Main approach.** The authors estimate a dynamic structural model of internal capital markets with multiple divisions, division productivity, costly investment adjustment, cash holdings, time-varying external financing costs, and managerial preferences for corporate socialism.
- **Dark-side estimate.** The baseline estimate of corporate socialism is $\lambda = 0.686$. In an average two-division firm, managers behave as if the stronger division is about 9% less productive and the weaker division is about 12% more productive than they really are.
- **Bright-side estimate.** At the median TED spread, the fixed cost of external financing is about 8.6% and the marginal cost is about 5% for the median two-division firm. A two-standard-deviation increase in TED roughly triples the fixed and marginal costs.
- **Key crisis result.** The model implies that internal capital markets offset financial-market stress during the crisis by about 16% for the average diversified firm and by about 30% for firms in the top quartile of productivity dispersion.
- **Economic takeaway.** Conglomerates are inefficient in normal times because of internal misallocation, but become relatively more valuable when external capital markets freeze because they can reallocate scarce internal funds.

2 Data and measurement

2.1 Compustat segment data and sample restrictions

- The main estimation data are Compustat segment files from 1980–2006. The counterfactual and validation exercise uses the 2007–2010 crisis period.
- Division-level variables include sales, assets, capital expenditures, operating profits, depreciation, and SIC codes.
- The paper excludes firms with incomplete segment information, financial and government divisions, sales below $10 million, missing market value or cash-flow data, and segment sales/assets that do not reconcile with firm-level totals.
- The resulting data include a large panel of diversified division-years; structural estimation focuses on conglomerates with two or three divisions, which cover about 90% of diversified firms in the sample.

Interpretation. The data are designed to observe both the allocation decision inside the firm and the division-level investment opportunities that headquarters is responding to.

2.2 Productivity and investment

- The baseline productivity measure is division **ROA**, defined as division cash flow divided by division assets.
- Division cash flow is operating profits plus reported accounting depreciation.
- Capital investment is capital expenditure normalized by assets.
- The paper also uses division industry Q as an alternative productivity measure for robustness.

Interpretation. ROA maps directly into the model because each division's cash flow is proportional to its capital and productivity. Industry Q is closer to the traditional conglomerate literature but is more exposed to measurement error.

2.3 Conglomerate value and productivity dispersion

- **Excess Value (EV)** is the log ratio of the diversified firm's value to the value of an imputed portfolio of comparable stand-alone firms.
- The comparable stand-alone firm for each division is chosen using the Lang and Stulz method.
- **Dispersion** is the standard deviation of imputed division market-to-book ratios within a diversified firm.
- In the descriptive statistics, the average diversified firm has EV of -0.108 and productivity dispersion of 0.423.

Interpretation. Dispersion is important because internal capital markets are most valuable when divisions have different investment opportunities, but also most vulnerable to corporate socialism when headquarters tilts toward weak divisions.

2.4 External capital market conditions

- The baseline external-market shifter is the **TED spread**, the spread between interbank lending rates and short-term U.S. Treasury bills.
- The paper interprets a high TED spread as impaired access to external capital.
- A robustness specification replaces TED with the Gilchrist-Zakrajsek excess bond premium (EBP).

Interpretation. The structural model needs a time-varying cost of external finance. TED and EBP provide aggregate financial-stress shifters that affect the relative value of internal capital markets.

3 Models and empirical specifications

3.1 Division productivity and capital accumulation

For division j at time t, productivity follows a Markov process:

$$z_{tj} = G_z(z_{t-1,j}) + \varepsilon_{ztj}. \tag{1}$$

Capital evolves according to

$$k_{tj} = (1-\delta)k_{t-1,j} + I_{t-1,j}. \tag{2}$$

Interpretation. Productivity determines the return to investing in a division. Capital is persistent because investment is irreversible and depreciates slowly.

3.2 Investment adjustment costs

The cost of investing in division j is

$$\Phi(I_{tj}, k_{tj}) = I_{tj} + \phi_0 \mathbf{1}_{I_{tj}>0} + \phi_1 k_{tj} \mathbf{1}_{I_{tj}>0} + \phi_2 \frac{I_{tj}^2}{k_{tj}}. \tag{3}$$

Interpretation. Investment is costly both because of fixed costs and because rapid adjustment is convexly costly. This prevents the model from attributing every investment difference to agency frictions.

3.3 Cash stock and external finance

Headquarters can finance investment using internal funds or by raising external finance f_t. Cash evolves as

$$p_{t+1} = p_t + f_t + \pi_t, \tag{4}$$

where π_t is firm profit. External capital-market conditions follow

$$\zeta_t = G_\zeta(\zeta_{t-1}) + \varepsilon_{\zeta t}. \tag{5}$$

The total cost of external finance is parameterized as a fixed and variable cost that depends on f_t, total assets, and ζ_t:

$$C(f_t \mid \zeta_t, \sum_j k_{tj}) = \mathbf{1}_{f_t>0}\left(c_0 + c_1\zeta_t + c_2\zeta_t^2 + c_3\frac{1}{\sum_j k_{tj}}\right) + \mathbf{1}_{f_t>0} f_t\left(c_4 + c_5\zeta_t + c_6\zeta_t^2 + c_7\frac{1}{\sum_j k_{tj}}\right) + \mathbf{1}_{f_t>0} c_8 f_t^2. \tag{6}$$

Interpretation. This is the model's bright side: when external finance is expensive, a conglomerate can economize on outside funds by moving cash internally across divisions.

3.4 Corporate socialism

Let z_t^* be the average productivity of the firm's divisions. The manager values division revenue as if division productivity were a weighted average of the division's true productivity and the firm's average productivity:

$$k_{tj}\left[(1-\lambda)z_{tj} + \lambda z_t^*\right]. \tag{7}$$

Equivalently, per-period utility can be written as

$$u_t = \sum_j \left[z_{tj}k_{tj} - \Phi(I_{tj}, k_{tj})\right] - C(f_t \mid \zeta_t, \sum_j k_{tj}) - L(p_t \mid \sum_j k_{tj}) - \lambda \sum_j (z_{tj} - z_t^*)k_{tj}. \tag{8}$$

Interpretation. If $\lambda > 0$, headquarters undervalues strong divisions and overvalues weak divisions. This captures the dark side of internal capital markets.

3.5 Dynamic problem and estimation

The manager chooses investment and financing policies to maximize discounted expected utility:

$$V(k_t, z_t, p_t, \zeta_t; \theta) = \max_{a_t} E_t \sum_{s=0}^{\infty} \beta^s u_{t+s}(k, z, p, \zeta; \theta), \tag{9}$$

where $a_t = (I_{t1}, \ldots, I_{tn}, f_t)$.

The paper estimates the model using the Bajari, Benkard, and Levin two-step estimator. The idea is:

1. estimate observed policy functions and state transitions;
2. simulate observed and alternative policies;
3. choose structural parameters so observed policies deliver weakly higher expected utility than feasible alternative policies.

Interpretation. The model is identified by how headquarters changes investment, financing, and cash policies when division productivity, productivity dispersion, cash, and external financial conditions change.

4 Identification and empirical design

4.1 What identifies the dark side

- The central variation is how a division's investment responds to its own productivity and to the productivity of other divisions in the same firm.
- If high-productivity divisions receive too little investment relative to low-productivity divisions, the model attributes part of that pattern to corporate socialism.
- The estimated λ quantifies how strongly headquarters pulls capital allocation toward equality of divisional cash flows.

Interpretation. Reduced-form regressions can show misallocation, but the structural model turns that misallocation into a quantitative preference parameter.

4.2 What identifies the bright side

- External financing costs are identified from the joint behavior of external financing, investment, and cash holdings.
- Time variation in TED or EBP shifts the cost of external finance.
- When external capital becomes expensive, internal capital markets should allocate capital more efficiently toward productive divisions.

Interpretation. The bright side is not just that conglomerates have more cash. It is that they can reallocate cash across divisions when stand-alone firms must raise costly outside funds.

4.3 Why the out-of-sample crisis test matters

- The structural parameters are estimated only through 2006.
- The model is then exposed to realized financial stress in 2008–2010.
- Firm value is not an input into estimation, but the simulated valuation pattern resembles actual crisis-period conglomerate valuations.

Interpretation. This is an external validation test. The model's ability to replicate crisis-period patterns makes it less likely that the estimates are driven solely by measurement error in productivity or by arbitrary structural assumptions.

4.4 Remaining limitations

- Productivity is difficult to measure at the division level. ROA and industry Q are imperfect proxies.
- Conglomerate composition may be endogenous; divisions may be grouped together because of unobserved synergies or selection.
- The counterfactual assumes the crisis operates only through external-finance costs, holding productivity and investment opportunities fixed.
- The model abstracts from the debt-equity choice and from an explicit stochastic discount factor.

Interpretation. The paper is strongest as a quantitative decomposition of costs and benefits of internal capital markets, not as a clean natural experiment.

5 Tables and figures

5.1 Figure 1 and Table 1: motivating facts and sample description

Read:

- Figure 1 plots excess value for high- and low-dispersion conglomerates together with TED. The valuation gap moves with financial stress, motivating a model in which external credit conditions alter the value of internal capital markets.
- Table 1 shows that conglomerate divisions are larger than stand-alone firms: mean sales are $756 million versus $494 million, and mean assets are $1.299 billion versus $768 million.
- Conglomerate divisions have higher ROA: 0.155 versus 0.116 for stand-alone firms.
- The average diversified firm has a negative excess value of -0.108, consistent with a conglomerate discount in normal times.

Economic message: The descriptive evidence motivates the paper's trade-off: diversified firms may be discounted on average, but financial stress can make their internal capital markets more valuable.

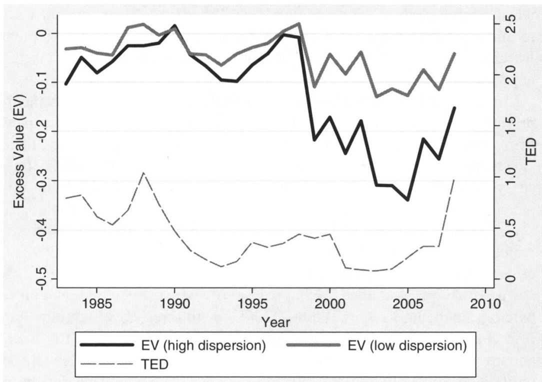

Figure 1: Excess value, dispersion, and TED

Table 1
Descriptive statistics

Sample: Manufacturing firms	Stand-alone		Conglomerate (division)	
	Mean	SD	Mean	SD
Sales (mm$)	494	2873	756	3723
Assets (mm$)	768	7162	1299	11661
Capex (mm$)	45.9	371	61.5	358
Industry Q	2.71	3.61	1.61	1.16
Capex/Assets	0.072	0.095	0.076	0.100
ROA	0.116	0.094	0.155	0.147
Excess Value (EV)			−0.108	0.514
Dispersion			0.423	0.764

The sample is by division and year (Compustat segment files, 1980–2006). Division sales, assets, capital expenditure, and cash flow are in millions of dollars. Capital investment (*Capex/Assets*) is measured as capital expenditure normalized by assets. *ROA* is defined as the cash flows of the division in that year divided by its assets. Division cash flow is defined as operating profits plus the reported accounting depreciation in the year. Industry Q of the division in a given year is the median Q of the stand-alone firms in the same industry. Excess Value (EV) of a diversified firm is calculated as the log of the ratio of firm value of a diversified firm relative to the portfolio of stand-alone firms, with the stand-alone firm corresponding to each division of the conglomerate chosen based on the method of Lang and Stulz (1994). Capital investment is measured as capital expenditure normalized by assets. *Dispersion* is defined as the standard deviation of the division (imputed) market-to-book ratio for a diversified firm.

Table 1: Descriptive statistics

5.2 Table 2 and Table 3: baseline structural estimates and alternative productivity measure

Read:

- Table 2 is the baseline structural estimation using ROA and TED. The corporate socialism parameter is $\lambda = 0.686$ with standard error 0.031.
- The baseline estimates imply strong internal misallocation: headquarters treats strong divisions as less productive and weak divisions as more productive than they are.
- At the median TED spread, the implied fixed external-financing cost is about 8.6%; the implied marginal cost is about 5%.
- The cost of holding cash is estimated around 28%, rationalizing why firms do not simply hoard unlimited internal cash.
- Table 3 repeats the structural estimation using industry Q as productivity. The corporate socialism estimate is $\lambda = 0.757$, close to the ROA result.

Economic message: The dark side is quantitatively large and robust across productivity measures. The paper's key structural object is not just a sign; it is an economically meaningful estimate of internal-capital-market distortion.

Table 2
Structural estimation: Estimates using *ROA* as a measure of productivity and TED as a measure of credit market conditions

$$\lambda\sum_j (z_{tj}-z^*)k_{tj}$$

λ
0.686
(0.031)

$$\Phi(I_{tj},k_{ij})=I_{tj}+\phi_0\mathbb{I}_{I_{tj}>0}+\phi_1\mathbb{I}_{I_{tj}>0}k_{tj}+\phi_2\frac{I_{tj}^2}{k_{tj}}$$

ϕ_0	ϕ_1	ϕ_2(x100)
8.721	0.003	0.01
(0.103)	(0.0001)	(0.001)

$$C(f_t|\zeta_t,\sum_j k_{tj})=\mathbb{I}_{f_t>0}(c_0+c_1\zeta_t+c_2\zeta_t^2+c_3\frac{1}{\sum_j k_{tj}})+\mathbb{I}_{f_t>0}(c_4+c_5\zeta_t+c_6\zeta_t^2+c_7\frac{1}{\sum_j k_{tj}})f_t+\mathbb{I}_{f_t>0}c_8 f_t^2$$

c_0	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8(x100)
63.32	−165.05	86.166	−30.149	0.36	−0.995	0.619	0.032	0.02
(0.998)	(1.440)	(0.763)	(0.761)	(0.002)	(0.008)	(0.011)	(0.001)	(0.001)

$$L(p_t|\sum_j k_{tj})=(l_0 p_t+l_1\frac{p_t}{\sum_j k_{tj}})$$

l_0	l_1
0.276	0.043
(0.001)	(0.0001)

The sample consists of nonfinancial, unregulated firms from Compustat segment files from 1980 to 2006. We report the estimated structural parameters as well as the standard errors on these estimates (in parentheses). We use *ROA* as a measure of productivity and use TED as a measure of external credit market conditions. The parameters map into the general utility function we discussed in the paper.

$$u_t=\underbrace{\sum_j (z_{tj}k_{tj}-\Phi(I_{tj},k_{tj}))}_{\text{production}}-\underbrace{\underbrace{C\left(f_t|\zeta_t,\sum_j k_{tj}\right)}_{\text{time varying cost of external finance}}\underbrace{-L\left(p_t|\sum_j k_{tj}\right)}_{\text{costof cash}}\underbrace{-\lambda\sum_j (z_{tj}-z^*)k_{tj}}_{\text{corporate socialism}}}_{\text{costly external finance}}$$

Table 2: ROA productivity and TED credit conditions

Table 3
Structural estimation: Estimates using *Q* as a measure of productivity and TED as a measure of credit market conditions

$$\lambda\sum_j (z_{tj}-z^*)k_{tj}$$

λ
0.757
(0.065)

$$\Phi(I_{tj},k_{ij})=I_{tj}+\phi_0\mathbb{I}_{I_{tj}>0}+\phi_1\mathbb{I}_{I_{tj}>0}k_{tj}+\phi_2\frac{I_{tj}^2}{k_{tj}}$$

ϕ_0	ϕ_1	ϕ_2
2.689	1.085	0.05
(1.004)	(0.009)	(0.001)

$$C(f_t|\zeta_t,\sum_j k_{tj})=\mathbb{I}_{f_t>0}(c_0+c_1\zeta_t+c_2\zeta_t^2+c_3\frac{1}{\sum_j k_{tj}})+\mathbb{I}_{f_t>0}(c_4+c_5\zeta_t+c_6\zeta_t^2+c_7\frac{1}{\sum_j k_{tj}})f_t+\mathbb{I}_{f_t>0}c_8 f_t^2$$

c_0	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8(x100)
−2.088	8.399	10.324	−85.989	0.689	−1.703	0.125	10.887	0.011
(1.139)	(1.915)	(5.422)	(2.734)	(0.273)	(0.106)	(0.086)	(0.204)	(0.103)

$$L(p_t|\sum_j k_{tj})=(l_0 p_t+l_1\frac{p_t}{\sum_j k_{tj}})$$

l_0	l_1
0.297	25.367
(0.054)	(2.358)

The sample consists of nonfinancial, unregulated firms from Compustat segment files from 1980 to 2006. We report the estimated structural parameters as well as the standard errors on these estimates (in parentheses). We use *Q* as a measure of productivity and use TED as a measure of external credit market conditions. The parameters map into the general utility function we discussed in the paper.

$$u_t=\underbrace{\sum_j (z_{tj}k_{tj}-\Phi(I_{tj},k_{tj}))}_{\text{production}}-\underbrace{\underbrace{C\left(f_t|\zeta_t,\sum_j k_{tj}\right)}_{\text{time varying cost of external finance}}\underbrace{-L\left(p_t|\sum_j k_{tj}\right)}_{\text{cost of cash}}\underbrace{-\lambda\sum_j (z_{tj}-z^*)k_{tj}}_{\text{corporate socialism}}}_{\text{costly external finance}}$$

Table 3: Q productivity and TED credit conditions

5.3 Table 4 and Table 5: robustness to EBP and alternative specifications

Read:

- Table 4 replaces TED with the excess bond premium. The corporate socialism parameter remains large: $\lambda = 0.651$.
- With EBP, the authors estimate a lower fixed cost of about 2% and a higher marginal cost of about 12%.
- Table 5 shows that λ remains close to the baseline across several parsimonious cost-function specifications: 0.687, 0.690, and 0.732.
- Only a very restrictive specification that removes fixed-cost financing terms and implies implausible marginal costs moves λ to 0.881.

Economic message: The main corporate-socialism estimate is not an artifact of the exact external-finance specification or of using TED rather than a bond-market stress measure.

Table 4
Structural estimation: Estimates using *ROA* as a measure of productivity and EBP as a measure of credit market conditions

$$\lambda \sum_j (z_{tj} - z^*) k_{tj}$$

λ
0.651
(0.042)

$$\Phi(I_{tj}, k_{ij}) = I_{tj} + \phi_0 \mathbb{I}_{I_{tj}>0} + \phi_1 \mathbb{I}_{I_{tj}>0} k_{tj} + \phi_2 \frac{I_{tj}^2}{k_{tj}}$$

ϕ_0	ϕ_1	ϕ_2 (x100)
12.210	0.006	0.030
(0.110)	(0.001)	(0.002)

$$C(f_t | \zeta_t, \sum_j k_{tj}) = \mathbb{I}_{f_t>0}(c_0 + c_1 \zeta_t + c_2 \zeta_t^2 + c_3 \frac{1}{\sum_j k_{tj}}) + \mathbb{I}_{f_t>0}(c_4 + c_5 \zeta_t + c_6 \zeta_t^2 + c_7 \frac{1}{\sum_j k_{tj}}) f_t + \mathbb{I}_{f_t>0} c_8 f_t^2$$

c_0	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8(x100)
8.641	−132.010	45.320	−42.510	0.091	0.202	0.177	0.397	0.040
(2.594)	(6.259)	(5.241)	(2.374)	(0.009)	(0.014)	(0.068)	(0.024)	(0.003)

$$L(p_t | \sum_j k_{tj}) = (l_0 p_t + l_1 \frac{p_t}{\sum_j k_{tj}})$$

l_0	l_1
0.397	0.061
(0.007)	(0.005)

The sample consists of nonfinancial, unregulated firms from Compustat segment files from 1980 to 2006. We report the estimated structural parameters as well as the standard errors on these estimates (in parentheses). We use *ROA* as a measure of productivity and use EBP (excess bond premium) as a measure of external credit market conditions. The parameters map into the general utility function we discussed in the paper.

$$u_t = \underbrace{\sum_j (z_{tj} k_{tj} - \Phi(I_{tj}, k_{ij}))}_{\text{production}} - \underbrace{\underbrace{C\left(f_t | \zeta_t, \sum_j k_{tj}\right)}_{\text{time varying cost of external finance}} - \underbrace{L\left(p_t | \sum_j k_{tj}\right)}_{\text{cost of cash}} - \underbrace{\lambda \sum_j (z_{tj} - z^*) k_{tj}}_{\text{corporate socialism}}}_{\text{costly external finance}}$$

Table 4: ROA productivity and EBP credit conditions

Table 5
Structural estimation: Robustness of corporate socialism parameter to alternative specifications

$$\lambda \sum_j (z_{tj} - z^*) k_{tj}$$

λ	Alternative specification
0.687 (0.030)	$c_8 = 0 \Leftrightarrow C(f_t \vert \zeta_t, \sum_j k_{tj}) = \mathbb{I}_{f_t>0}(c_0 + c_1 \zeta_t + c_2 \zeta_t^2 + c_3 \frac{1}{\sum_j k_{tj}}) + \mathbb{I}_{f_t>0}(c_4 + c_5 \zeta_t + c_6 \zeta_t^2 + c_7 \frac{1}{\sum_j k_{tj}}) f_t$
0.690 (0.032)	$c_2, c_6, c_8 = 0 \Leftrightarrow C(f_t \vert \zeta_t, \sum_j k_{tj}) = \mathbb{I}_{f_t>0}(c_0 + c_1 \zeta_t + c_3 \frac{1}{\sum_j k_{tj}}) + \mathbb{I}_{f_t>0}(c_4 + c_5 \zeta_t + c_7 \frac{1}{\sum_j k_{tj}}) f_t$
0.732 (0.044)	$c_2, c_6, c_8, c_3, c_7, l_1 = 0 \Leftrightarrow C(f_t \vert \zeta_t, \sum_j k_{tj}) = \mathbb{I}_{f_t>0}(c_0 + c_1 \zeta_t) + \mathbb{I}_{f_t>0}(c_4 + c_5 \zeta_t) f_t$ and $L(p_t \vert \sum_j k_{tj}) = (l_0 p_t)$
0.881 (0.052)	$c_0, c_1, c_2, c_6, c_8, c_3, c_7, l_1 = 0 \Leftrightarrow C(f_t \vert \zeta_t, \sum_j k_{tj}) = \mathbb{I}_{f_t>0}(c_4 + c_5 \zeta_t) f_t$ and $L(p_t \vert \sum_j k_{tj}) = (l_0 p_t)$

The sample consists of nonfinancial, unregulated firms from Compustat segment files from 1980 to 2006. We report the estimated structural parameters as well as the standard errors on these estimates (in parentheses). We use *ROA* as a measure of productivity and use TED as a measure of external credit market condition. The parameters map into the general utility function we discussed in the paper.

$$u_t = \underbrace{\sum_j (z_{tj} k_{tj} - \Phi(I_{tj}, k_{ij}))}_{\text{production}} - \underbrace{\underbrace{C\left(f_t | \zeta_t, \sum_j k_{tj}\right)}_{\text{time varying cost of external finance}} - \underbrace{L\left(p_t | \sum_j k_{tj}\right)}_{\text{cost of cash}} - \underbrace{\lambda \sum_j (z_{tj} - z^*) k_{tj}}_{\text{corporate socialism}}}_{\text{costly external finance}}$$

Table 5: Robustness of λ

5.4 Figure 2 and Table 6: counterfactual crisis simulation and actual crisis data

Read:

- Figure 2 shows the out-of-sample counterfactual. When TED spikes in 2008, simulated excess value improves for diversified firms relative to stand-alone firms.
- The same figure shows that high-Q divisions receive more investment during the credit shock, while low-Q divisions receive less investment relative to comparable stand-alone firms.
- Table 6 Panel A reports the simulated-data regressions. During years 1–3, dispersion has a positive effect on EV (0.013), and diversified-firm investment becomes more sensitive to Q (coefficient on $Q\times$ diversified $= 0.232$).
- In the simulation, high-dispersion conglomerates have a stronger investment-Q sensitivity than low-dispersion conglomerates: 0.278 versus 0.181.
- Table 6 Panel B shows actual crisis data from 2007–2009. The interaction between dispersion and the 2008/2009 crisis dummy is positive (0.053), and investment sensitivity also rises for diversified firms.

Economic message: The counterfactual and actual data tell the same story: external market stress makes internal capital markets more efficient, especially for conglomerates with dispersed investment opportunities.

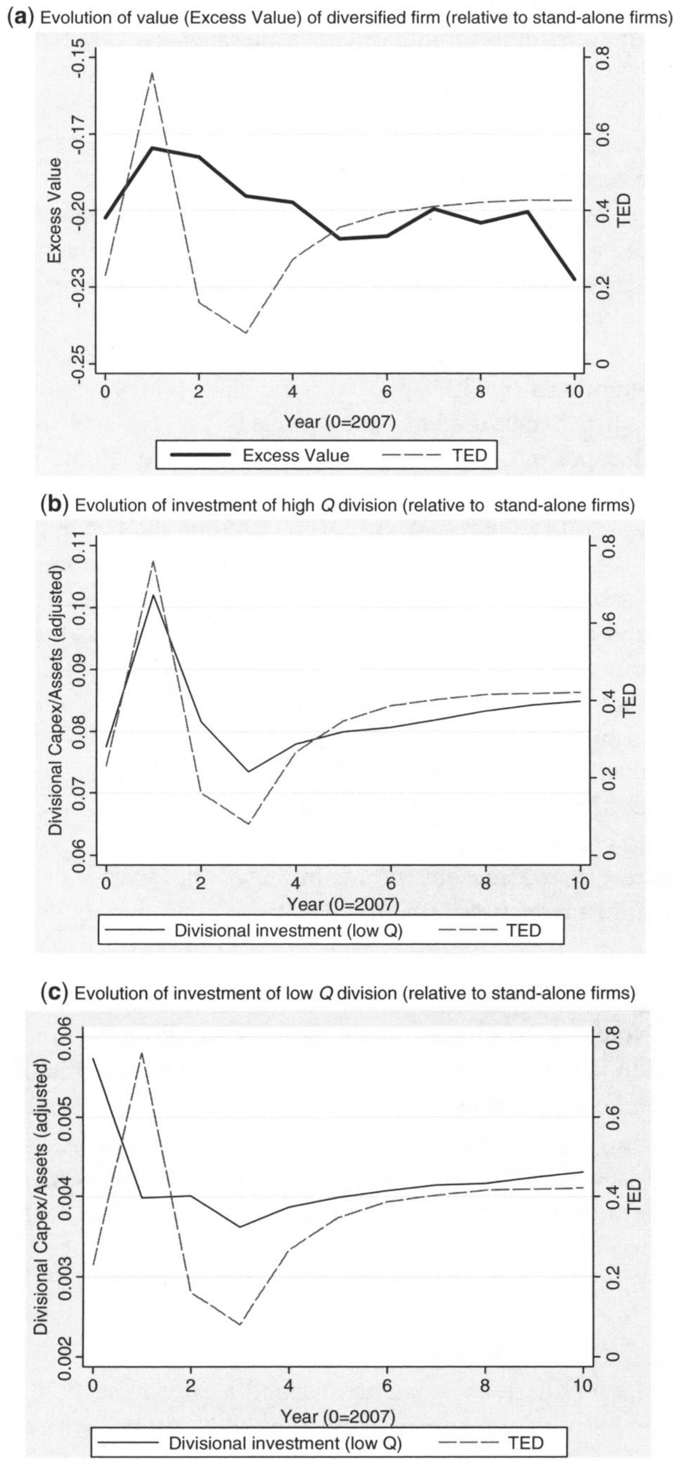

Figure 2: Counterfactual evolution of EV and divisional investment

Table 6
Out-of-sample test (counterfactual): Excess value and capital expenditures in response to the crisis

Panel A: Data from forward simulation

	EV	*Capex/Assets*	*Capex/Assets*	*Capex/Assets*	*Capex/Assets*	EV
	Sample: Diversified firms (year 1–3)	Sample: All firms (year 1–3)	Sample: Diversified firms (year 1–3)	Sample: Diversified firms (year 1–3)		Sample: Diversified firms (postcrisis: year 4–6)
				High Dispersion	*Low Dispersion*	
	(1)	(2)	(3)	(4)	(5)	(6)
Q		–0.001	0.231***	0.278***	0.181***	
		(0.002)	(0.019)	(0.031)	(0.019)	
Q*Dummy$_{(Diversified)}$		0.232***				
		(0.006)				
Dispersion	0.013**					–0.112***
	(0.005)					(0.015)
Observations	792	5940	1584	824	760	792
R-squared	0.052	0.410	0.111	0.148	0.094	0.044
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Year 0 = 2007						

Panel B: Actual data from Compustat (2007–2009)

	EV	*Capex/Assets*	*Capex/Assets*	*Capex/Assets*	*Capex/Assets*
	Sample: Diversified firms	Sample: All firms	Sample: Diversified firms	Sample: Diversified firms	
				High Dispersion	*Low Dispersion*
	(1)	(2)	(3)	(4)	(5)
Q		−0.005***	−0.001	−0.001*	0.002
		(0.001)	(0.001)	(0.0006)	(0.003)
Q*Dummy$_{(Diversified)}$		0.007***			
		(0.001)			
Q X Dummy$_{(Year=08 \text{ or } 09)}$			0.002**	0.003***	−0.006
			(0.001)	(0.001)	(0.004)
Dispersion	−0.105***				
	(0.0273)				
Dispersion X Dummy$_{(Year=08 \text{ or } 09)}$	0.053***				
	(0.016)				
Dummy$_{(Year=08 \text{ or } 09)}$	0.0240*		−0.0140***	−0.0147***	−0.002
	(0.013)		(0.002)	(0.003)	(0.007)
Observations	5837	21408	11674	6677	4997
R-squared	0.03	0.04	0.04	0.04	0.04
Other controls	Yes	Yes	Yes	Yes	Yes

The table reports regressions based on the counterfactual exercise. Panel A presents results from data that use estimates based *only* on the data from 1980 to 2006. We start with a random sample of conglomerate and stand-alone firms at the end of 2007 and expose the firms in the sample to realized values of our shifter of capital market conditions, TED, for 2008, 2009, and 2010. TED spread is the difference between the interest rates on interbank loans and short-term U.S. government Treasury bills. We forward simulate our model (based on parameters in Table 3) with simulations of productivity shocks from 2008 and shocks to TED from 2010 for 100 periods. The dependent variables used in the regressions are Excess Value (EV) and capital expenditure normalized by assets (*Capex/Assets*). Panel B uses the same dependent variables and presents results using actual data from Compustat segment files for the period 2007, 2008, and 2009. Dummy$_{(Diversified)}$ is an indicator variable that takes a value 1 if the firm has more than one segment, and *High (Low) Dispersion* are all diversified firms that have above

Table 6: Simulated and actual crisis regressions

6 Short summary

- The paper studies whether internal capital markets can offset external capital-market dislocations.

- It estimates a structural model in which diversified firms have both agency costs and financing advantages.
- The agency cost is corporate socialism: managers tilt investment toward weaker divisions.
- The estimated socialism parameter is large: $\lambda \approx 0.69$ in the baseline, and remains close under alternative productivity and credit-stress measures.
- The financing benefit is that diversified firms can reallocate internal funds when external finance is expensive.
- The 2007–2010 counterfactual shows that financial stress makes conglomerates relatively more valuable because they can shift capital toward better divisions.
- The bright side partially offsets the crisis shock by 16% to 30%, depending on the level of productivity dispersion.
- The key lesson is state dependence: the value of firm boundaries depends on the condition of external capital markets.

7 Conclusion

7.1 Core conclusion

Internal capital markets are not always good or always bad. In normal times, corporate socialism causes inefficient capital allocation. In crisis times, the ability to move scarce funds internally offsets external-financing frictions and raises the relative value of diversified firms.

7.2 Economic intuition

A stand-alone firm with a good project must raise outside funds when its internal funds are insufficient. When credit markets are stressed, this is expensive. A conglomerate can instead move funds from one division to another. This advantage is especially valuable when divisions have very different productivity. The cost is that headquarters may also keep weak divisions alive by overallocating capital to them.

7.3 Takeaway

The paper's main contribution is a quantitative decomposition of the dark and bright sides of internal capital markets. Firm boundaries matter for macro-financial shocks because some firms can internally reallocate resources when the external financial system is impaired.